Guide to River Extraction and Sectioning from Satellite Images

Introduction

The software uploaded in this Zenodo repository has been adopted to generate the set of results in the research paper entitled:

- S. Rizzello et al., Multiscaling Behaviour of Braided Channel Networks: A New Formulation of the Taylor's Law Variate Transformations (DOI to be updated).

In particular, the code allows to extract the river domain and its sectioning statistics from satellite raw images, in BMP file type (minor changes would be needed to import from a different image format).

The software to extract the river domain is named *river.extractor.nb*, whereas the sectioning algorithm is named *sectioning.nb*. Both the codes are implemented in Wolfram Mathematica language. The main view of the code *river.extractor.nb* is shown in Figure 1, whereas the main view of the code *sectioning.nb* is reported in Figure 2.

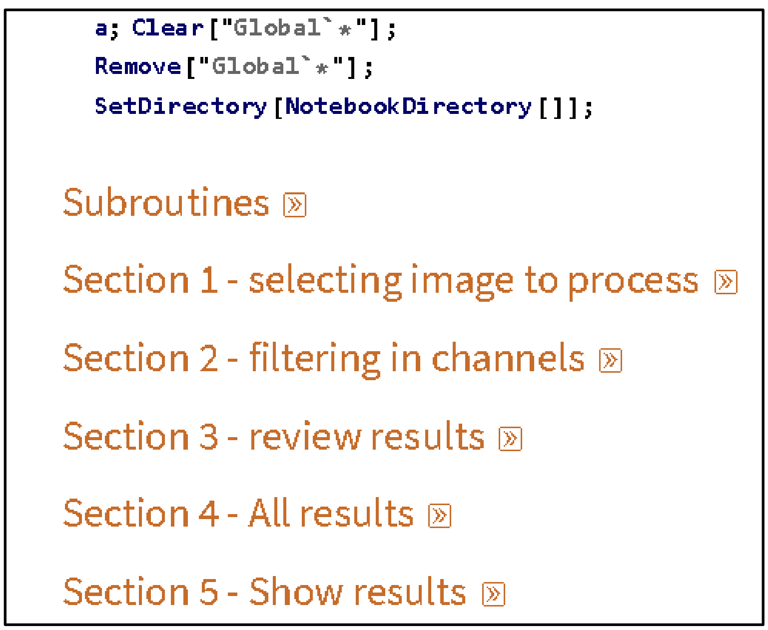

Figure 1 – Main view of river.extractor.nb.

```
a = 1; Clear["Global`*"];
Remove["Global`*"];
SetDirectory[NotebookDirectory[]];

Subroutines
Select image
Fit river gross section with spline: select mesh size
Extract crossing sections with 1px resolution
Post-processing
```

Figure 2 – Main view of sectioning.nb.

Usage of *river.extractor.nb*

The code is divided in 5 sections, which need to be executed one by one. In particular:

1) Section 1: Select image to process.

Images in BMP formats need to be available in the same folder where the *river.extractor.nb* is located. A list of available BMP images is produced by the code, and the variable **imagenumber** need to be populated with the ordering number of the image to analyse.

If the specific image has not been already processed before, the software will ask to execute Sec. 2, otherwise it will suggest executing Sec. 3.

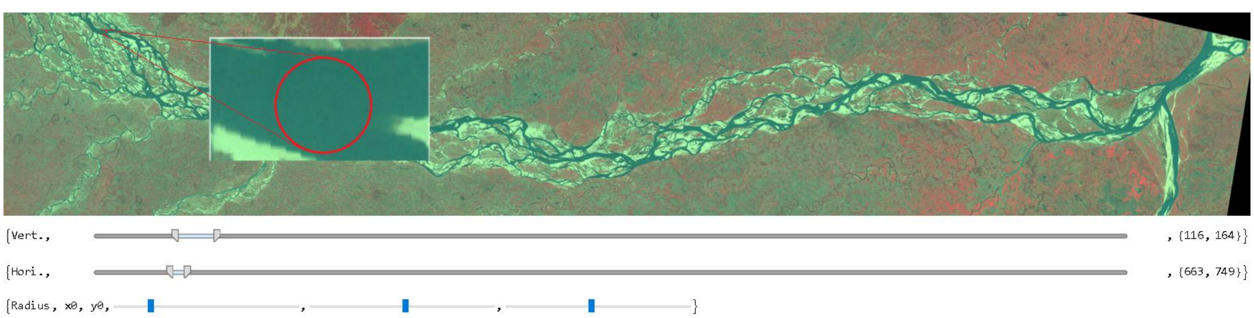

Figure 3 - Select and magnify portion of the river to extract river colours statistics.

2) Section 2: Get colour statistics from the channels.

This part is devoted to collect colour statistics from few river samples, in order to identify the full river domain in the whole satellite image. Since the river can be characterized by different colour levels through all its extension, the suggestion is to acquire at least three river samples, equally distributed along the image. To do this, the magnification bars and the circle (identifying the area to be analysed), need to be displaced manually as desired. Figure 3 shows an example of application, where the circled area to be extracted is searched on the very left side of the river image. The following code paragraph needs to be executed ("extractPDFwater") in order to retrieve the colour statistics, fitted with joint Gaussian probability density function. The previous circle search and statistics extraction need to be executed in other parts of the river domain -as suggested before-, in order to improve the river colour statistics.

Once this statistics population execution has been performed, the river can be extracted from the satellite image by executing the paragraph "stddistance". In particular, the variable `stddistance` needs to be populated with a real positive number, and it describes the prefactor to the standard deviation adopted when considering an image point to belong to the river.

3) Section 3: Review the just analysed satellite image

This section is used to review the river extraction results, by comparing extracted river image with satellite image. The extracted river image can thus be saved in the "extracted" folder, containing all the extracted rivers.

4) Section 4: Loads all the analysed satellite images available in the folder.
5) Section 5: Makes video with a sequence of the analysed river images.

Usage of *sectioning.nb*

This code is adopted for the sectioning of the river extracted with the previous algorithm. The river images will be searched in the "extracted" folder.

In particular, after selecting the image to analyse, the algorithm will fit the river with a spline. This river fit is done in order to perform the river sectioning, thus the following cross-section statistics extraction, perpendicularly to the local spline position. Indeed, the river shows a long wave fluctuation, which is removed thanks to the spline fitting.

The image to process is selected as previously in the Section "Select image". Once this has been done, the whole Mathematica page can be executed thoroughly. In the Section "Post-processing", the relevant statistics are derived and plotted.